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LAB REPORT

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Remarks

Uncertainty-Aware Adversarial Defense for Security-Critical Intrusion Detection

Pipeline

1. Dataset preparation
2. Baseline: Random Forest, XGBoost, MLP
3. Adversarial robustness evaluation
4. Uncertainty estimation + selective prediction
5. Defense comparison + statistical validation

কাজের ধাপ

- Dataset: CIC-IDS2017 অথবা UNSW-NB15।
- Normalization, class imbalance handling এবং leakage-free train/validation/test split।
- Clean input-এ baseline performance নির্ণয়; এরপর অনুমোদিত threat model অনুযায়ী adversarial perturbation পরীক্ষা।
- Uncertainty calibration, abstention threshold এবং robustness-aware training তৈরি।
- Clean accuracy, adversarial performance, attack success rate, calibration error, coverage-risk curve ও inference overhead তুলনা।

মূল novelty পরীক্ষা: একই robustness অর্জনে কম computational cost হয় কি না এবং uncertainty-based abstention বাস্তবে ভুল security decision কমায় কি না।

Title: Uncertainty-Aware Adversarial Defense for Security-Critical Intrusion Detection

Abstract

Intrusion Detection Systems (IDS) play an important role in network security, but machine learning-based models can be vulnerable to adversarial attacks. This project investigates adversarial robustness and uncertainty-aware prediction using the UNSW-NB15 dataset. Random Forest, XGBoost, and Multi-Layer Perceptron (MLP) were implemented as baseline models. FGSM was used to evaluate adversarial vulnerability, while confidence-based uncertainty estimation and selective prediction were explored. Adversarial training strategies were also compared. Experimental results show that adversarial attacks significantly reduce model performance, while the implemented defense did not consistently improve robustness. The project highlights the importance of evaluating robustness, prediction reliability, and computational efficiency together.

1. Introduction

Machine learning-based intrusion detection systems help identify malicious network activities. However, adversarial perturbations can manipulate model predictions and reduce detection reliability.

This project investigates how adversarial attacks affect intrusion detection models and whether uncertainty-aware prediction and adversarial training can improve their reliability.

2. Objectives

- Build baseline intrusion detection models.
- Evaluate model performance on clean and adversarial data.
- Investigate uncertainty estimation and selective prediction.
- Compare adversarial training strategies.
- Analyze the trade-off between robustness and computational cost.

3. Dataset and Preprocessing

The UNSW-NB15 dataset was used for binary intrusion detection.

- Training dataset: 175,341 records.
- Testing dataset: 82,332 records.
- Features: 34 input features.
- Target: Normal (0) and Attack (1).
- The attack_cat column was excluded to prevent target leakage.

Preprocessing included missing-value handling, categorical encoding, and numerical scaling for MLP. The original training dataset was divided into training and validation sets using an 80:20 stratified split. The original test set was reserved for final evaluation.

Duplicate records and train-test overlap were also examined and documented as potential evaluation limitations.

4. Baseline Models and Results

Three machine learning models were implemented:

- Random Forest
- XGBoost
- Multi-Layer Perceptron (MLP)

The models were evaluated on the original test dataset.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Forest	86.63%	81.90%	97.20%	88.90%	97.61%
XGBoost	86.86%	81.91%	97.73%	89.12%	98.09%
MLP	85.19%	80.45%	96.58%	87.78%	97.13%

XGBoost achieved the best overall baseline performance. However, its normal-class recall was relatively low, indicating a considerable number of false positive predictions.

5. Adversarial Robustness Evaluation

The Fast Gradient Sign Method (FGSM) was used to generate adversarial examples against the MLP model.

Perturbations were applied to standardized numerical features while categorical one-hot features were kept unchanged.

Epsilon	Accuracy	Recall	F1-score
0.00	85.19%	96.58%	87.78%
0.01	80.15%	93.49%	83.84%
0.05	66.67%	80.55%	72.69%
0.10	44.75%	57.79%	53.53%
0.20	30.66%	42.70%	40.41%

The results demonstrate that increasing adversarial perturbation strength significantly degrades model performance.

6. Uncertainty Estimation and Selective Prediction

A confidence-based uncertainty score was calculated using the model's predicted probabilities:

Uncertainty = 1 - |2p - 1|

A rejection threshold of 0.2 was selected using validation data. Predictions with higher uncertainty were rejected for potential secondary inspection.

Final test results:

- Coverage: 74.64%
- Accepted samples: 61,450
- Rejected samples: 20,882
- Selective Accuracy: 97.83%
- Selective Precision: 96.86%
- Selective Recall: 99.81%
- Selective F1-score: 98.31%

These metrics apply only to accepted samples and should not be directly compared with overall baseline accuracy. Rejected samples require further inspection.

7. Adversarial Defense and Statistical Validation

Two adversarial training approaches were explored:

1. Single-Epsilon Adversarial Training.
2. Multi-Epsilon Adversarial Training.

The multi-epsilon approach used adversarial examples generated with different perturbation strengths. However, the defense did not outperform the original MLP under the evaluated test attacks.

Model	**Epsilon**	**Accuracy**	**F1-score**
Original MLP	0.00	85.19%	87.78%
Original MLP	0.05	66.67%	72.69%
Original MLP	0.10	44.75%	53.53%
Original MLP	0.20	30.66%	40.41%
Multi-Epsilon Defense	0.00	81.87%	85.65%
Multi-Epsilon Defense	0.05	28.81%	41.44%
Multi-Epsilon Defense	0.10	25.61%	39.85%
Multi-Epsilon Defense	0.20	25.01%	39.63%

A paired McNemar test was conducted on validation predictions to compare the original MLP and multi-epsilon defense. The result was statistically significant ($p = 1.05 \times 10^{-9}$), indicating a difference in their clean prediction correctness. The original MLP performed better on clean validation data.

8. Limitations

- The adversarial evaluation used constrained numerical-feature FGSM rather than realistic packet-level manipulation.
- The uncertainty score was confidence-based and not formally calibrated.
- Temperature Scaling and PGD-based defense were not implemented.
- Duplicate records and train-test overlap may affect generalization estimates.
- The implemented adversarial defense did not consistently improve robustness.

9. Conclusion and Future Work

This project evaluated machine learning-based intrusion detection under adversarial conditions using UNSW-NB15. XGBoost achieved the best baseline test performance, while FGSM experiments revealed significant vulnerability in the MLP model. Confidence-based selective prediction improved accuracy among accepted samples, but the explored adversarial training strategy did not provide consistent robustness improvements.

Future work will investigate PGD-based adversarial training, probability calibration, duplicate-aware data splitting, realistic feature constraints, and computationally efficient defense strategies.

5 AI suggestions Pipeline

ChatGPT:

Proposed Improved Research Pipeline

Research Title: Uncertainty-Aware and Compute-Efficient Adversarial Defense for Security-Critical Intrusion Detection

Stage 1: Dataset Preparation and Leakage Analysis

- Use the UNSW-NB15 dataset.
- Analyze duplicate records, conflicting labels, and train-test overlap.
- Apply leakage-aware train-validation-test splitting.
- Perform categorical encoding, numerical scaling, and class distribution analysis.

Stage 2: Baseline Modeling

- Train Random Forest, XGBoost, and MLP models.

- Evaluate Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC.
- Measure training time, inference latency, and memory usage.

Stage 3: Adversarial Robustness Evaluation

- Define an explicit threat model and perturbation constraints.
- Evaluate FGSM and PGD attacks where feasible.
- Compare clean and adversarial performance.
- Measure Attack Success Rate and false positive/negative rates.

Stage 4: Uncertainty Estimation and Calibration

- Establish confidence-based uncertainty estimation.
- Apply Temperature Scaling to improve probability calibration.
- Evaluate Expected Calibration Error (ECE) and Brier Score.
- Compare uncertainty under clean and adversarial conditions.

Stage 5: Selective Prediction and Risk-Aware Abstention

- Determine rejection thresholds using validation data.
- Route high-uncertainty predictions for secondary inspection.
- Evaluate prediction coverage, selective risk, and coverage-risk curves.
- Analyze whether abstention reduces incorrect security decisions.

Stage 6: Compute-Efficient Adversarial Defense

- Compare standard training, FGSM adversarial training, and multi-epsilon adversarial training.
- Investigate PGD adversarial training where computationally feasible.
- Compare clean accuracy, adversarial robustness, training cost, and inference overhead.

Stage 7: Statistical Validation and Final Comparison

- Perform paired statistical tests where appropriate.
- Report confidence intervals for key metrics.
- Conduct ablation studies to evaluate individual components.
- Compare robustness, calibration, selective risk, and computational efficiency.

Expected Research Contribution

The proposed framework aims to investigate whether uncertainty-aware abstention and computationally efficient adversarial training can improve the reliability of intrusion detection systems under adversarial conditions.

Research Questions:

1. Can adversarial robustness be improved without excessive computational overhead?
2. Can uncertainty-based abstention reduce incorrect security decisions under adversarial attacks?
3. What trade-offs exist among robustness, prediction coverage, calibration, and computational cost?

Implementation Status: The baseline models, FGSM evaluation, confidence-based uncertainty estimation, selective prediction, adversarial training experiments, and validation-based McNemar test have been completed. PGD evaluation, Temperature Scaling, and further leakage-aware evaluation remain future work.

Claude

Uncertainty-Aware Adversarial Defense for Security-Critical Intrusion Detection

Pipeline

1. **Data:** CIC-IDS2017 + UNSW-NB15. Remove duplicates, temporal/flow-grouped split, fit scaler and resampling on train only.
2. **Baselines:** Random Forest, XGBoost, MLP (+ one tabular deep model such as FT-Transformer). Report Macro-F1 and per-class recall.
3. **Threat model:** White-box and black-box attacks (PGD, HopSkipJump) with feature-constrained perturbations (only attacker-controllable features). Report ASR and robust accuracy.
4. **Robust training:** Standard vs PGD-AT vs Fast AT vs TRADES. Measure training time and inference latency.
5. **Uncertainty**: MC-Dropout, small ensemble, temperature scaling, conformal prediction. Report ECE, coverage-risk curve, and test abstention under adaptive attack.
6. **Validation:** 5 seeds, mean ± std, Wilcoxon test, ablation study.

Novelty Questions

- Same robustness at lower cost (Fast AT/TRADES vs PGD-AT)?
- Does abstention reduce wrong security decisions, even under adaptive attack?

Why Better

- Realistic constrained attacks
- Two datasets for generalization
- No data leakage
- Uncertainty tested under attack

- Cost claims backed by numbers
- Statistical rigor

Gemini

Research Proposal: Uncertainty-Aware Adversarial Defense for Security-Critical Intrusion Detection

1. **Problem & Threat Model Definition**
    - Feature Bounds: Categorize features into Immutable (IP/Protocol), Bounded (Packet Length), and Perturbable.
    - Domain Constraints: Enforce perturbation limits so generated adversarial packets remain protocol-valid.
2. **Preprocessing & Data Splitting**
    - Session/Temporal Split: Split CIC-IDS2017 / UNSW-NB15 chronologically to prevent train-test data leakage.
    - Imbalance Handling: Apply cost-sensitive loss weighting instead of SMOTE.
3. **Model Architecture & Uncertainty**
    - Baselines: XGBoost / LightGBM, TabNet / MLP.
    - Uncertainty Modeling: Implement MC-Dropout / Ensembles to separate Epistemic (attack/OOD) from Aleatoric (noise) uncertainty.
4. **Domain-Constrained Defense Evaluation**
    - Constrained Attacks: Test under feature-bounded PGD and AutoAttack variants.
    - Adaptive Abstention & Fallback: Route low-confidence samples to a rule-based/DPI checker to stop Abstention DoS attacks.
5. **Metrics & Statistical Validation**
    - Performance: Clean Accuracy, Robust Accuracy, Attack Success Rate (ASR).
    - Uncertainty & Overhead: Calibration Error (ECE), Latency per sample (ms), Risk-Coverage AUC.
    - Significance Testing: Validate improvements using Wilcoxon Signed-Rank Test.

Pipeline Enhancements vs Baseline

1. **Attack Realism**

- Baseline: Unconstrained PGD/FGSM.
- Proposed: Domain-Constrained Perturbations.
- Rationale: Prevents invalid packets that real routers drop.

2. Uncertainty Reliability

- Baseline: Softmax Confidence.
- Proposed: Epistemic vs Aleatoric Disentanglement.
- Rationale: Softmax becomes overconfident on adversarial samples.

3. Operational Safety

- **Baseline:** Static Abstention Threshold.
- **Proposed:** Dynamic Abstention + DPI Fallback.
- **Rationale:** Avoids Denial-of-Service vulnerabilities from forced abstentions.

4. Evaluation Rigor

- **Baseline:** Random Split & Simple Runtime.
- **Proposed:** Temporal Split & Pareto Latency-Robustness Curve.
- **Rationale:** Prevents temporal data leakage and proves real-world efficiency.

QWEN AI

Proposed Paper Direction: Evidential & Cost-Aware Adversarial Defense for Security-Critical Intrusion Detection

1. Data & Threat Modeling

- Datasets: CIC-IDS2017 or UNSW-NB15.
- Preprocessing: Leakage-free, time-series-aware splits; handle class imbalance.
- Threat Model: Feature-space attacks (modifying valid network features like packet size/flow duration), not unrealistic pixel-level perturbations.

2. Baseline & Attack Generation

- Models: Lightweight differentiable models (1D-CNN, MLP, or TabNet) instead of tree-based models.
- Attacks: Constrained white-box (PGD, CW) and black-box attacks ensuring perturbed features remain network-valid.

3. Core Novelty: Single-Pass Uncertainty & Selective Defense

- Uncertainty: Evidential Deep Learning (EDL) for single-pass uncertainty estimation (avoids multi-pass latency of MC Dropout).
- Selective Prediction: Abstain on high-uncertainty inputs.
- Fallback: Route abstained traffic to a lightweight rule-based system/DPI to prevent missed attacks.

4. Rigorous Evaluation

- Performance: Clean Accuracy, Adversarial Accuracy, Attack Success Rate (ASR).
- Uncertainty: Expected Calibration Error (ECE), Coverage-Risk Curve.
- Security: False Positive Rate (FPR) at 95% TPR, Cost-Sensitive Metrics.
- Efficiency: Inference latency (ms), FLOPs, and memory footprint.

Why This Pipeline is Better

1. Fixes the Gradient Problem: Replaces Random Forest/XGBoost with differentiable models, enabling actual gradient-based adversarial training and evaluation.
2. Proves "Low Computational Cost": EDL calculates uncertainty in a single forward pass, unlike MC Dropout or Ensembles which require 10-20 passes, directly validating your efficiency novelty.
3. Realistic Attacks: Uses feature-space constraints, making the adversarial examples practically deployable in real networks.
4. Actionable Abstention: Adds a fallback mechanism, ensuring that when the model says "I don't know," the system still makes a safe security decision instead of just failing.
5. Security-Aligned Metrics: Evaluates FPR@95% TPR and cost-sensitivity, which matters more in cybersecurity than raw accuracy.

Perplexity AI

Title:
Constrained Threat-Aware Selective Defense for Adversarially Robust Intrusion Detection

1. **Dataset Preparation**
    - Use UNSW-NB15 and CIC-IDS2017.
    - Perform leakage-free normalization and stratified train/validation/test splitting.
    - Handle class imbalance using class weights or balanced sampling.
2. **Baseline Models**
    - Train Random Forest, XGBoost, and MLP.
    - Evaluate clean performance using accuracy, macro-F1, recall, and false positive rate.
3. **Adversarial Evaluation**
    - Apply FGSM, PGD, and black-box transfer attacks.
    - Use feature-constrained perturbations to make the threat model realistic.
    - Measure robust accuracy, attack success rate, and false alarm rate.

4. **Uncertainty and Selective Prediction**
    - Estimate uncertainty using MC Dropout, Deep Ensembles, or evidential deep learning.
    - Apply calibration using temperature scaling.
    - Create three decisions: accept, flag for review, and conservative action.
    - Tune the abstention threshold using expected security cost.

5. Defense Comparison

- Compare with no defense, adversarial training, defensive distillation, and uncertainty-only abstention.
- Run all experiments with at least five random seeds.
- Use McNemar's test or bootstrap confidence intervals for validation.

6. Evaluation Metrics

- Clean accuracy and macro-F1
- Robust accuracy
- Attack success rate
- False alarm rate
- Calibration error
- Coverage-risk curve
- Inference latency and computational cost

Main Novelty

The main contribution is a **cost-sensitive, uncertainty-based selective defense** that reduces incorrect security decisions under realistic adversarial attacks while maintaining lower computational cost than full adversarial training.

Key Research Question

Can uncertainty-based abstention reduce missed attacks and false alarms under adversarial perturbation without significantly increasing computational overhead?